WAN – WIDE AREA NETWORK

✓ Default Cisco Router Serial Encapsulation is HDLC- (High level data link controller)

sh int s0

✓ Default Cisco Router Ethernet encapsulation is ARPA- (Advanced research project agencies)

sh int fa0

PPP authentication: (Point to Point Protocol)

The calling side of the link must transmit information to ensure that the sender is authorized to establish the connection. This is accomplished by the series of authentication message between the Routers. PPP supports two types of authentication:

PAP: Password Authentication Protocol. (supports 1 way and 2-way Authentication)

CHAP: Challenge handshake authentication protocol. (supports three-way Authentication)

RA

```
# Configure Terminal
# username trichy password 123
# int fa0
# ip add 10.0.0.1 255.0.0.0
# No sh
# int s0
# encapsulation ppp
# ip add 20.0.0.1 255.0.0.0
# no sh
# router ospf 100
# network 10.0.0.0   0.255.255.255  area 0
# network 20.0.0.0   0.255.255.255  area 0
# Int s0
# ppp pap sent-username Coimbatore password abc
# ppp authentication pap
```

RB

```
# Configure Terminal
# username coimbatore password abc
# int fa0
# ip add 30.0.0.1 255.0.0.0
# No sh
# int s0
# encapsulation ppp
# ip add 20.0.0.2 255.0.0.0
# no sh
# router ospf 100
# network 20.0.0.0   0.255.255.255   area 0
# network 30.0.0.0   0.255.255.255   area 0
# Int s0
# ppp pap sent-username trichy password 123
# ppp authentication pap
```

RA & RB

```
# debug ppp authentication
```

RA or RB

```
# int s0
# sh
# no sh
```

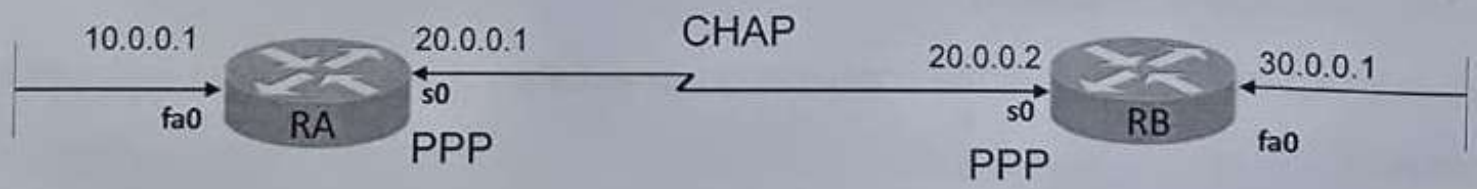
CHAP (Challenge Handshake Authentication Protocol)

CHAP uses a three-way handshake to force remote hosts to identify themselves after the link establishments phase. CHAP performs the following steps after the link establishment phase is complete:

- ✓ The local router that received the call sends a challenge packet to the remote host that initiated the call. the challenge packet consists of an ID, a random number, and either the name of the local host performing the authentication or a username on the remote host.
- ✓ The remote host must respond with its encrypted unique ID, a one – way encrypted password , the remote hostname or a username, and a random number.
- ✓ The local router performs its own calculation on the response values. It accepts or rejects the authentication request based on whether the value it received from the remote host matches the value it calculated.
- ✓ Like PAP, CHAP terminates the connection immediately if the local host rejects the authentication.

CHAP (Challange Handshake Authentication Protocol)

Password Must Be Same



Hardware & Networking Academy (P) Ltd | Robotics & IOT Solutions | Web Designing SEO & Digital Marketing Solutions | Mechatronics & AI Solutions | Software Technologies

TRICHY
Aruvi Block, St. Paul's Complex,Bharathiyar Salai, Cantonment, Trichy - 620 001
75022 02555

COIMBATORE
Ajay Building, 9th Street, Cross Cut Road, Gandhipuram, Coimbatore - 641 012
75022 02555

CHENNAI
Thilakavathy Complex, Mudichur Road, Krishna Nagar, West Tambaram, Chennai- 600 045
75022 02555

contact@systechgroup.in
systechgroup.in
systechgroup

RA

```
# Configure Terminal
# hostname trichy
# int fa0
# ip add 10.0.0.1 255.0.0.0
# No sh
# int s0
# encapsulation ppp
# ip add 20.0.0.1 255.0.0.0
# no sh
# router ospf 100
# network 10.0.0.0  0.255.255.255  area 0
# network 20.0.0.0  0.255.255.255  area 0
```

RB

```
# Configure Terminal
# hostname coimbatore
# int fa0
# ip add 30.0.0.1 255.0.0.0
# No sh
# int s0
# encapsulation ppp
# ip add 20.0.0.2 255.0.0.0
# no sh
# router ospf 100
# network 20.0.0.0  0.255.255.255  area 0
# network 30.0.0.0  0.255.255.255  area 0
```

CHAP

```
# configure terminal
# username coimbatore password 123
# int s0
# ppp authentication chap
```

```
# configure terminal
# username trichy password 123
# int s0
# ppp authentication chap
```

RA & RB

```
# debug ppp authentication
```

RA or RB

```
# int s0
# sh
# no sh
```

Hardware & Networking Academy (P) Ltd | Robotics & IOT Solutions | Web Designing SEO & Digital Marketing Solutions | Mechatronics & AI Solutions | Software Technologies

TRICHY
Aruvi Block, St. Paul's Complex, Bharathiyar Salai, Cantonment, Trichy - 620 001
75022 02555
contact@systechgroup.in

COIMBATORE
Ajay Building, 9th Street, Cross Cut Road, Gandhipuram, Coimbatore - 641 012
75022 02555
systechgroup.in

CHENNAI
Thilakavathy Complex, Mudichur Road, Krishna Nagar, West Tambaram, Chennai- 600 045
75022 02555
systechgroup